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Author(s): Robert L. Smith

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Efficient Monte Carlo Procedures for Generating Points Uniformly Distributed over Bounded Regions

ROBERT L. SMITH
The University of Michigan, Ann Arbor, Michigan
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We consider the Monte Carlo problem of generating points uniformly distributed within an arbitrary bounded (measurable) region. The class of Markovian methods considered generate points asymptotically uniformly distributed within the region. Computational experience suggests the methods are potentially superior to conventional rejection techniques for large dimensional regions.

WE CONSIDER the following problem. Given a bounded k-dimensional surface $S \subset R^n$ where $k \leq n$, how can we efficiently generate a pseudorandom point $(X_1, X_2, \cdots, X_n) \in S$ uniformly distributed over S. That is, we require that the probability $X \in A \subset S$ be given by $V(A)/V(S)$ where V is the k-dimensional content of (Jordan) measurable sets in S. The problem is actually quite general in that the seemingly broader problem of generating points X distributed over a region S according to a probability density function f over S is in fact equivalent. To see this equivalence, merely generate points Y uniformly distributed within the region under the graph of f over S and project Y onto S yielding X. X will have the desired properties.

Such a procedure is of interest for several reasons. As noted above, the procedure could be used in generating multivariate random deviates with arbitrary density functions. Efficient transformational procedures are currently limited to the multivariate normal and a few other specialized distributions (Rubinstein [1981], Schmeiser [1981]). Another potential application is in generating large numbers of feasible solutions to mathematical programs uniformly distributed within or on the boundary of the feasible region. These may serve as random starting points for heuristic optimum seeking procedures (Dixon and Szego [1978]) or may serve by themselves as stochastic probes yielding statistical information

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1. BACKGROUND

Conventional approaches to generating uniformly distributed points over bounded regions S are limited to transformational, rejection, and composition techniques. We consider each in turn together with their relative merits.

Transformation techniques directly capitalize on the fact that extremely efficient pseudorandom number generators are widely available for producing sequences of real numbers drawn from the interval [0, 1] that have the statistical properties of being independent and identically distributed over their range (Hammersley and Handscomb [1964]). A set of k such numbers $X = (X_1, X_2, \cdots X_k)$ regarded as a point in R^k yields a pseudorandom point uniformly distributed within the standard k-dimensional hypercube, H. A *transformation technique* then maps X via a smooth deterministic function T onto S. A necessary and sufficient condition that $T(x)$ be uniformly distributed over S is that the Jacobian of T be constant over all $x \in H$ and in fact be equal to $V(S)$. Generating a uniform distribution on the unit hypersphere by scaling a multivariate normal vector of independent components can be viewed as a composition map of a transformation of points in a unit cube to normals followed by a cross-section transformation. However, the resulting transformation T being nonsquare has no determinant in this case. This technique is extremely efficient when T is easily computed. Unfortunately, T is simple and known only for a limited class of regions S. For example, for paralleletopes T is linear, for hyperspheres T is a simple closed form trigonometric function, and for simplices T is a weighted linear combination of the extreme points of S (Patel and Smith [1979], Rubinstein [1982]). Unfortunately, the latter idea does not generalize to polytopes. We could, of course, partition any bounded polytope into a finite union of simplices and use transformational techniques on the simplices (an example of a *composition* technique (Schmeiser)). Although conceptually sound, the number of simplices and the difficulty of identifying them makes the procedure computationally intractable for even moderately large dimensional problems. Hsuan [1979] suggests this method for polygons where the dimensionality is low enough for the procedure to be efficient.

Rejection techniques (Hammersley and Handscomb; Rubinstein [1982]) rely on the simple principle that if a point is uniformly distributed within a region D enclosing a region S, then it is conditionally uniformly distributed within S given it lies in S. The procedure encloses S by a region D for which a transformation technique is known. Using the transformation T to generate a point $T(X)$ in D, we accept $T(X)$ if it also lies in S and is otherwise rejected. Each point so accepted is then

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uniformly distributed within S. Typical choices for D include a rectangular paralleletope, a hypersphere, and a simplex. For example, if the region S were a bounded polytope, finding D as a paralletope would involve ascertaining maximal feasible ranges on the coordinate variables. Selecting D as a simplex or sphere would require solving a linear or quadratic program, respectively. Depending on the complexity of S, one or more of these options may be computationally infeasible. A more serious problem of rejection techniques is that the number of trial points in D that must be generated to obtain a point in S grows explosively in the dimension of the region S. For example, when S is a hypercube and the enclosing region D is a circumscribed hypersphere, the expected number of points that must be generated within D to obtain one within S grows from 1.5 for dimension $k = 2$ to 10^{30} for $k = 100$. The n-dimensional content for an n-dimensional sphere of radius r is $2r^n\pi^{n/2} \cdot (n\Gamma(n/2))^{-1}$ (Kendell [1961]). Hence the content of the sphere enclosing an n-dimensional unit cube is given by $2(n\pi/4)^{n/2}(n\Gamma(n/2))^{-1}$.

In the next section we consider a class of procedures that recursively generates a sequence of points all within the region S. These points have the property that when the initial point is uniformly distributed within S then all of them are. Moreover, the last point generated is asymptotically independent of the first point as the sequence grows in length. In short, if we generate a long sequence of points and randomly mix their order, the resulting sequence of points has the statistical properties of being independent and identically distributed points uniformly distributed over S.

2. SYMMETRIC MIXING ALGORITHMS AND THEIR ASYMPTOTIC PROPERTIES

We define here a class of randomized algorithms that have the property of (strong) *reversibility*. That is, the probability density of moving from point x to point y in a region S is equal to the probability density of going from y to x. Consequently, were the algorithms run backward in time, their probabilistic properties would remain unaltered and it would be impossible to distinguish which endpoint of the generated path was the origin. We will show that this property implies a uniform stationary distribution over the points so generated. With a stronger assumption requiring that all subregions be eventually visited, we will further conclude that the points generated asymptotically approach a uniform distribution over the region regardless of the starting point.

The class of algorithms to be considered is subsumed by the following general mixing algorithm. The character of a particular algorithm is imparted through a choice of the direction set D as described below.

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on the value of the optimal objective function (Patel and Smith [1983], de Haan [1981]).

Mixing Algorithm

1. Choose a starting point $X_0 \in S$ and set $i = 0$.
2. Generate a random direction d uniformly distributed over a *direction set* $D \subseteq R^n$. Find the *line set* $L = S \cap \{x \mid x = x_0 + \lambda d, \lambda \text{ a real scalar}\}$ and generate a random point X_{i+1} uniformly distributed over L.
3. If $i = N$, stop. Otherwise, set $i = i + 1$ and return to 2.

Boneh and Golan [1979] first proposed using such an algorithm with D a hypersphere to identify nonredundant constraints in nonlinear programs. They also conjectured, but did not prove, that the points generated were uniformly distributed. Smith [1980] later, but independently, proposed the same algorithm and demonstrated its asymptotic uniformity.

A *symmetric mixing algorithm* is a mixing algorithm that is reversible in the sense described above. If S is an open set in R^n, every mixing algorithm is symmetric since the probability of traveling from x to a neighborhood of y is, by construction, the same as traveling from y to a neighborhood of x. Before cataloging the assumptions we will later need to impose on the class of symmetric mixing algorithms considered, we will first note that the points generated by a general mixing algorithm $X_0, X_1, \cdots$ constitute a continuous state Markov chain for $k > 0$ and a discrete state Markov chain for $k = 0$. Moreover, the chain is homogeneous. Let $P(A \mid x) = P(X_{i+1} \in A \mid X_i = x)$ for measurable $A \subseteq S$ be the one step transition probability. We will later impose one or both of the following regularity conditions on $P(A \mid x)$ and, by implication, on the class of mixing algorithms and regions considered. Roughly speaking, the conditions require the algorithm to distribute points in a continuous fashion over the entire region S.

Assumptions. (a) There is a measurable function

$$f(y \mid x) \text{ on } S \times S \text{ such that}$$

$$P(A \mid x) = \int_A f(y \mid x) dy \text{ for all measurable } A \subseteq S.$$

(b) $f(y \mid x) > 0$ for all $x, y \in S$.

Note that for the class of symmetric mixing algorithms that satisfy Assumption (a), the transition probability density function f is symmetric, i.e., $f(y \mid x) = f(x \mid y)$ for all $x, y \in S$. Roughly speaking, the probability of going from x to y is the same as that of going from y to x since they share the same line direction and the successor point is chosen uniformly along the shared line segment.

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LEMMA 1. *Under Assumption (a), the uniform distribution over S is a stationary initial distribution for the Markov chain induced by a symmetric mixing algorithm, i.e.,* $\lambda(A) = \int_S P(A \mid x)\, \lambda(dx)$ *where* $\lambda(A) = V(A)/V(S)$ *with A measurable and V representing k-dimensional content over S.*

Proof. For a symmetric mixing algorithm, $f(y \mid x) = f(x \mid y)$ and, therefore,

$$\int_S P(A \mid x)\lambda(dx) = \int_S \int_A f(y \mid x)\lambda(dy)\lambda(dx) = \int_A \int_S f(x \mid y)\lambda(dx)\lambda(dy)$$
$$= \int_A P(S \mid y)\lambda(dy) = \int_A \lambda(dy) = \lambda(A).$$

The interchange of order of integration is justified by Fubini's theorem since $f(y \mid x) \geq 0$ and the double integral is finite.

Lemma 1 states that if the starting point X_0 were chosen uniformly from S, then X_i would remain uniformly distributed over S for all $i \geq 0$.

LEMMA 2. *Let $X_0, X_1, X_2, \cdots$ be the Markov chain induced by a symmetric mixing algorithm. Under Assumptions (a) and (b), the chain is indecomposable, i.e., there are no two disjoint closed sets $A_1, A_2 \subset S$.*

A nonempty set A is *closed* if it is almost surely not possible to leave it, i.e., $P(A \mid x) = 1$ for all $x \in A$.

Proof. Suppose A_1 and A_2 are closed and $A_1 \cap A_2 = \phi$. Set $\bar{A} = S - (A_1 \cup A_2)$. Since $A_1, A_2, \bar{A}$ form a partition of S, $\lambda(A_1) + \lambda(A_2) + \lambda(\bar{A}) = 1$, and hence one of them has strictly positive measure, say A_1. Since A_2 is nonempty, choose $x \in A_2$. Then $P(A_1 \mid x) = \int_{A_1} f(y \mid x)\, dy > 0$ since $f > 0$. Hence, $P\,(A_2 \mid x) < 1$ which is a contradiction. Suppose now $\lambda(\bar{A}) > 0$. By the same argument, $P(A_2 \mid x) < 1$ for $x \in A_2$ and we obtain a contradiction. The result follows.

THEOREM 1. *Let $X_0, X_1, X_2, \cdots$ be the Markov chain induced by a symmetric mixing algorithm. Under Assumption (a) and (b), the uniform distribution λ is the unique stationary distribution over S, and moreover, the Markov chain $X_0, X_1, X_2, \cdots$ with initial distribution λ is ergodic, i.e.,* $\lim_{m\to\infty}(1/m) \sum_{i=0}^{m} \chi_A(X_i) = \lambda(A)$ *for measurable $A \subseteq S$ where*

$$\chi_A(x) = \begin{cases} 1 & \text{for} \quad x \in A \\ 0 & \text{for} \quad x \notin A \end{cases}.$$

Proof. From Lemma 2, we have that $X_0, X_1, X_2, \cdots$ is indecomposable and from Lemma 1 that λ is a stationary initial distribution. As an immediate consequence (see Proposition 7.11, p. 134, and Theorem 7.16, p. 136, in Breiman [1968]), λ is the unique stationary distribution and

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the process with λ as its initial distribution is stationary and in fact ergodic.

In this proof and throughout our discussion, we refer to Breiman even though his results are stated and proven for measurable $S \subseteq R^1$. The extension to R^n is only notational since the ordering properties of R^1 are not necessary to any of the results we use here.

Theorem 1 states that the sequence of points generated by a symmetric mixing algorithm satisfying Assumptions (a) and (b) will visit each subregion of S in the limit a fraction of time equal to that region's fractional content. That is, the points $X_0, X_1, X_2, \cdots$ considered as an ensemble will be uniformly distributed in the long run throughout the region S. Put another way, a Chi-square frequency test would be unable to statistically distinguish the points $X_0, X_1, X_2, \cdots$ generated by the symmetric mixing algorithm from those generated by transformational or rejection techniques; i.e., their observed long run frequencies of occurrence would be in agreement with an independent and identically distributed uniform model for $X_0, X_1, X_2, \cdots$.

THEOREM 2. *Under Assumption (a) and (b), the Markov chain $X_0, X_1, X_2, \cdots$ induced by a symmetric mixing algorithm is strongly mixing, i.e., $lim_{m\to\infty} P(X_m \in A \mid X_0 = x) = \lambda(A)$ for all starting points $x \in S$ and for all measurable sets $A \subseteq S$.*

Proof. Doob [1948] has shown that if a stationary initial distribution exists, then the chain is strongly mixing under the following conditions: (i) the state space S is indecomposable under $P(A \mid x)$; (ii) the motion is nonperiodic, i.e., S is indecomposable under the n-step transitions probability $P^{(n)}(A \mid x) n = 2, 3, \cdots$; and (iii) for each $x \in S$, $P(A \mid x)$ is absolutely continuous with respect to the stationary initial distribution. In our case, a stationary initial distribution exists by Lemma 1 and (i) follows from Lemma 2. (ii) follows from the same argument used in establishing Lemma 2 since $f^{(n)}(y \mid x) = \int_S f^{(n-1)}(z \mid x) f(y \mid z) dz > 0$ by induction on n. As for (iii), $P(A \mid x)$ is a bounded indefinite integral of a nonnegative measurable function $f(y \mid x)$ and all indefinite integrals of integrable functions are absolutely continuous with respect to the defining measure of integration (Theorem II, p. 97, Halmos [1950]).

Theorem 2 implies that the serial correlation between any two points X_k and X_{k+M} goes to zero as the number of iterations separating them M becomes large. This conclusion follows since X_k and X_{k+M} are asymptotically independent in the limit as M goes to infinity. Consequently, shuffling (randomly permuting) the indices in a large number of generated points $X_0, X_1, \cdots, X_M$ to produce $X_{(0)}, X_{(1)}, \cdots, X_{(M)}$ will induce the same uncorrelated effect for *any* pair of distinct points $X_{(k)}$ and $X_{(k')}$.

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That is, the shuffled sequence $X_{(0)}, X_{(1)}, \cdots, X_{(M)}$ will not only exhibit frequencies in accordance with a uniform model, but will also exhibit a lack of serial correlation in agreement with an independent model. In short, the statistical properties of $X_{(0)}, X_{(1)}, \cdots, X_{(M)}$ will for large M approximate those of M i.i.d. uniform random deviates. As a result, all statistics that are invariant under permutations of the data used in their calculation (e.g., a minimum or an average) may be based directly on the original sequence $X_0, X_1, \cdots, X_M$ but will nonetheless have the statistical properties suggested by an i.i.d. uniform model for $X_0, X_1, \cdots, X_M$. A very important issue in all this is of course how large M must be. We will explore this issue in the last section of the paper.

We will now consider specific algorithms corresponding to different choices of the direction set D all of which are symmetric algorithms that moreover satisfy Assumptions (a) and (b) for a large class of regions S.

A Random Directions Algorithm (Boneh and Golan [1979], Smith [1980]):

Set $D = \{d \in R^n \mid \|d\| = 1\}$. Then selecting d uniformly over the unit sphere D is the operational equivalent of selecting a random direction. Such a d is easily computed by many different techniques. Perhaps the simplest is to generate n independent normally distributed random deviates $N = (N_1, N_2, \cdots, N_n)$ and set $d = N/\|N\|$ (see e.g., Knuth [1969, p. 116]).

Establishing the most general class of regions S for which Assumptions (a) and (b) hold for this algorithm is a difficult task that we will not attempt here. However, the next section establishes Assumptions (a) and (b) for all open regions $S \subseteq R^n$. In particular, S is assumed to be of full dimension n. Arguments for lower dimensional surfaces are not difficult if S is specialized further. For example, if S is the surface of a n-dimensional polytope, it can be argued directly that $P(A \mid x)$ can be written as an integral of a positive probability density function over S. However, we lose symmetry unless the algorithm is altered to stall at the current point favoring movement in directions that meet facets less orthogonally. Such a random "bouncing" over the facets of S will by Theorems 1 and 2 be asymptotically uniformly distributed over S. The implied search procedure may have merit for ill-structured, but small dimensional mathematical programs.

A Coordinate Directions Algorithm (J. Telgen [private communication]):

Set $D = \{d \in R^n \mid d = \pm e^i$ for some $i = 1, 2, \cdots, n\}$, where $e_j^i = 1$ for $j = i$ and 0 otherwise$\}$. Although this algorithm was independently suggested by several people (including this author), Telgen first noted

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the algorithm's considerable computational advantage over the Random Directions Algorithm on a per iteration basis. The difference is particularly compelling in the important case where $S = \{x: Ax \leq b\}$. As noted by Telgen, the full sparsity of A may be exploited in every iteration in determining the end points of the line segment L. On the other hand, the number of iterations necessary to converge to the uniform distribution may offset this advantage, although preliminary computational experience suggests the converse.

Establishing Assumptions (a) and (b) for this algorithm is complicated by the fact that $P(A \mid x)$ is typically not absolutely continuous with respect to the uniform distribution over S. The probability mass is concentrated along the coordinate axes which are of zero Lebesque measure for open regions S. However, if the infimum of all line set lengths over open S is positive, then every open subregion may be reached in a finite number of iterations M_0. It can then be argued that the M_0-step transition probability $P^{(M_0)}(A \mid x)$ satisfies Assumptions (a) and (b). It follows that the sequence of points $X_0, X_{M_0}, X_{2M_0}, \cdots$ is asymptotically uniform over such regions S.

Actually, a stronger claim may be made if we approximate S by a rectangular mesh oriented along a rectangular coordinate system. Such an approximation has no practical restriction since computer accuracy is finite. The coordinate directions algorithm now generates a finite state Markov chain of points $X_0, X_1, X_2, \cdots$ where the number of states N is the number of rectangular parallelepipeds in the mesh. The corresponding transition probability matrix (P_{ij}) is symmetric, irreducible, and aperiodic. Therefore, (P_{ij}) is doubly stochastic, and, hence (Ross [1983]), $\lim_{m\to\infty} P_{ij}^m = 1/N$ for all $i, j = 1, 2, \cdots, N$.

A Vertex Reaching Algorithm:

To keep the discussion simple, suppose P is an n-dimensional simple polytope with vertices $V_1, V_2, \cdots, V_r$ so that each vertex has exactly n neighboring vertices. Let $S = \{V_1, V_2, \cdots, V_r\}$ and think of P transformed so that the direction set $D = \{d_1, d_2, \cdots, d_n\}$ can be constructed of directions that point to the n adjacent vertices regardless of the current vertex V_i. Then the dimension k of S is zero, and the k-dimensional content reduces to the cardinality of subsets of S. Hence $\lambda(V_i) = 1/r$ for all i and clearly $f(y \mid x) = 1$ for all $x, y \in S$. From Theorems 1 and 2, the procedure of starting at any vertex of P and randomly moving along edges to adjacent vertices in an unbiased way will asymptotically "lose" one among the vertices of P. Since the vertices visited are in the long run probabilistically equivalent to i.i.d. uniform choices from S, the distribution of the number of vertices visited until a distinguished vertex is visited (say the optimum vertex in a concave minimization problem)

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approximately follows a geometric distribution. In particular, the expected number of vertices visited before success is the number of vertices r. This analysis is not intended to recommend a search procedure, but rather to illustrate how Theorems 1 and 2 can aid in the probabilistic analysis of the general class of randomized algorithms we are considering.

3. CONVERGENCE RATES FOR THE RANDOM DIRECTIONS ALGORITHM

The practical merit of the symmetric mixing algorithms is of course dependent on their rate of convergence to the uniform distribution. We have explicitly calculated upper bounds on the rate of convergence to uniform for the random directions algorithm and the results are reported in this section. The formula for the bound gives considerable insight on what problem characteristics are important in determining convergence rate. However, the bound is of computational interest only for low dimensional problems. As we shall see below, it can overestimate the number of iterations required to achieve uniformity by several orders of magnitude.

THEOREM 3. *Let $X_0, X_1, X_2, \cdots$ be the Markov chain generated by the Random Directions Algorithm over an open region $S \subset R^n$. Then for any measurable set $A \subseteq S$, $|P(X_m \in A \mid X_0 = x) - \lambda(A)| < (1 - (\gamma/n2^{n-1}))^{m-1}$ where γ is the ratio of the n-dimensional content of S to the n-dimensional content of the smallest sphere containing S.*

Proof. From Case (b), p. 197, in Doob [1953] $|P(X_m \in A \mid X_0 = x) - \lambda(A)| \leq (1 - \delta\phi(C))^{m-1}$ where ϕ is a finite measure on a σ-field over S and C is a measurable set in S for which $\phi(C) > 0$ and $f(y \mid x) \geq \delta > 0$ for all $x, y \in C$ where $f(y \mid x)$ is the density of the absolutely continuous component of P with respect to ϕ. In our case, ϕ is n-dimensional content V. To obtain the strongest bound, we set C equal to S. Our task then is to show $f(y \mid x)$ exists, and hence that P is itself absolutely continuous, and to find $\delta > 0$. $f(y \mid x) = \lim_{V(A)\to 0} P(A \mid x)/V(A)$ if this limit exists for $y \in A$. Take A as an n-dimensional cube of content l^n oriented along the ray from x to y. For l small, $P(A \mid x) \approx (2l^{n-1}/S_n(r(x, y)))(l/d(x, y))$ where $r(x, y)$ is the distance from x to y, $d(x, y)$ is the diameter of S along the ray from x to y, and $S_n(x)$ is the surface area of an n-dimensional sphere of radius x. Hence $f(y \mid x) = \lim_{l\to 0} P(A \mid x)/l^n = 2/S_n(r(x, y))d(x, y)$ exists. Moreover, $r(x, y) \leq d$ and $d(x, y) \leq d$ where $d = \max_{x,y\in S} d(x, y)$. Hence, $f(y \mid x) > \delta = 2/dS_n(d)$ for all $x, y \in S$. Since $S_n(d) = n2^n V_n(d/2)/d$ where $V_n(x)$ is the volume of an n-dimensional sphere of radius x (Kendall [1961, p. 35]), we get $\delta = 1/n2^{n-1}V_n(d/2)$. But $V_n(d/2)$ is the volume of a circumscribed sphere around S and, hence, the result follows since $\phi(C) = V(S)$.

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As a numerical illustration of Theorem 3, consider the case when S is a disk in R^2. Then $|P(X_m \in A \mid X_0 = x) - \lambda(A)| \leq (3/4)^{m-1}$. The convergence as in all cases is exponentially fast. For example, we will be within 1% of a uniform distribution over S after 17 iterations of the random directions algorithm regardless of starting point. Clearly, some starting points X_0 are more conducive to rapid convergence than others, and for this as well as other reasons, 17 iterations is only an upper bound on the required number.

It is interesting that the ratio γ of the content of S to an enclosing sphere plays a role here in algorithm efficiency just as in the standard rejection technique. However, a large number of iterations to achieve uniformity in the random directions algorithm case also implies a correspondingly large number of uniform points corresponding to each of the iterations. However, the rejection technique retains only the last (feasible) point.

Of course, the practical merit of the approach rests on its efficiency for large dimensional problems. The Random Directions Algorithm was therefore run on a 10-dimensional hypercube region with an initial point randomly chosen according to a uniform distribution. We chose this regular shape in order to easily test for uniformity of the iteration points generated. Note that a cube is the easiest of all regions to generate uniform points within by conventional approaches. However, the measure of difficulty for the Random Directions Algorithms is the difference between the region generated and an enclosing sphere. The ratio of their corresponding contents in this case is roughly 250 to 1. This result gives a theoretical upper bound of over five million iterations to get within an ϵ of 1% of uniformity. We, however, ran the algorithm for only 10,000 iterations. To avoid serial correlation effects, we first sampled every 10th point generated. We then shuffled this smaller population of 1,000 points to randomize the ordering. Note that the points were shuffled, but not their components. We then performed chi-square frequency and serial correlation tests (Knuth, p. 35) on this new sequence of 1,000 points. Two-tailed chi-square tests were performed since true distributional uniformity would have an observable variance component in a departure from deterministic uniformity. The cells corresponded to 10 equal volume slabs in the X_i direction for $i = 1, 2, \cdots, 10$, which correspond to 10 chi-square tests in all. Note that the X_i direction is a favored direction only with respect to the shape of the region and not the performance of the algorithm which is blind to coordinate orientations. Hence, aspects of multivariate as opposed to only marginal uniformity are being tested. Tables I and II report the results.

The results are consistent with the hypothesis that the points generated are independent and identically distributed uniformly over the cube,

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TABLE I

CHI-SQUARE FREQUENCY TEST RESULTS FOR 1000 POINTS GENERATED IN A 10-DIMENSIONAL CUBE[a]

i	Observed Frequency f_{ij} of Points in Slab j in Coordinate Direction X_i Out of a Total of 1,000 Points										Frequency χ^2 Statistic ($\nu = 9$ *d.f.*)	Passed Uniformity at $\alpha = 10\%$ Level of Significance
	j											
	1	2	3	4	5	6	7	8	9	10		
1	88	98	86	82	132	104	102	104	116	88	21.3	No
2	97	122	99	105	94	94	94	99	96	100	6.4	Yes
3	81	84	103	99	107	106	91	117	104	108	11.6	Yes
4	111	112	124	122	110	89	85	83	85	79	27.2	No
5	115	107	98	106	97	93	94	93	92	105	5.5	Yes
6	123	99	91	102	106	99	93	91	92	104	8.6	Yes
7	106	84	106	92	82	96	115	94	108	117	13.5	Yes
8	95	94	110	91	107	74	88	121	114	106	17.8	No
9	94	91	98	103	94	91	116	111	105	97	6.6	Yes
10	116	100	92	99	95	99	115	102	90	92	7.4	Yes

[a] Upper and lower χ^2 values for $\alpha = 10\%$, $\nu = 9$: (3.3, 16.9).

although 3 failures of 10 for the frequency test for an α of 10% is statistically significant. This result may, however, be due to the lack of independence among the 10 chi-square tests. The 10,000 iterations required is two orders of magnitude less than would be predicted by the upper bound of Theorem 3. Were the region a slight distortion from a perfect cube (e.g., with spikes coming out of the centers of its faces to touch an enclosing sphere), we could still expect essentially the same performance of the Random Directions Algorithm. However, a conventional rejection technique of enclosing the region in a sphere, say, would

TABLE II

CHI-SQUARE SERIAL TEST RESULTS FOR 1,000 POINTS GENERATED IN A 10-DIMENSIONAL CUBE[a]

Coordinate Direction X_i	Serial χ^2 Statistic ($\nu = 99$ *d.f.*)	Passed Serial Correlation at $\alpha = 10\%$ Level of Significance
1	107.6	Yes
2	85.6	Yes
3	105.6	Yes
4	111.6	Yes
5	84.4	Yes
6	96.0	Yes
7	125.6	No
8	103.6	Yes
9	82.4	Yes
10	103.6	Yes

[a] Upper and lower χ^2 values for $\alpha = 10\%$, $\nu = 99$: (77.9, 124.3).

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result in only one generated point in roughly 250 iterations. Indeed, finding the enclosing sphere itself may be computationally intractable. It is evident that the mixing algorithm approach can be a very significant computational improvement over conventional Monte Carlo techniques when dealing with irregularly shaped but large-dimensional regions.

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